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Software Requirements Specification (SRS)

Hospital Management System

Subject: Software Engineering
Assignment 01

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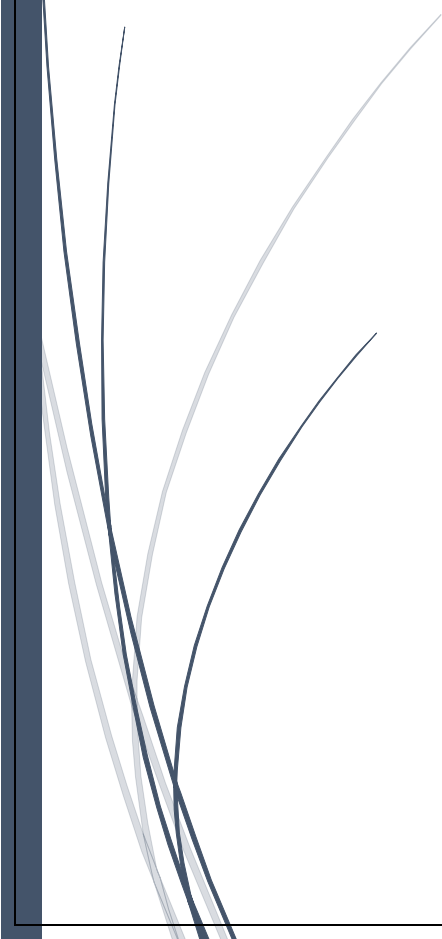


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1)Introduction

1.1)Purpose of the system

The purpose of this Software Requirements Specification (SRS) document is to define the functional and non-functional requirements of the Hospital Management System (HMS). This document serves as a guideline for developers, stakeholders and users to understand the system's behavior, features and constraints.

The HMS is designed to improve hospital operations by digitizing patient management, appointment scheduling, billing and medical record handling.

1.2)Scope

The Hospital Management System (HMS) is a web-based application developed to streamline and automate hospital operations. The system will provide functionalities such as,

- ✓ Patient registration and management
- ✓ Doctor scheduling and management
- ✓ Appointment booking and tracking
- ✓ Medical record management
- ✓ Billing and payment processing
- ✓ Inventory management
- ✓ Reporting and analytics

The system will be used by hospital staff, doctors, administrators and patients to improve efficiency and reduce manual errors.

1.3)Definitions and abbreviations

- HMS – Hospital Management System
- Admin – System Administrator
- OPD – Outpatient Department
- DB – Database
- UI – User Interface
- API – Application Programming Interface

2)Overall Description

2.1)Product perspective

The Hospital Management System is a standalone web-based application developed using modern technologies,

- ✓ Frontend: React.js
- ✓ Backend: Node.js with Express.js
- ✓ Database: MongoDB

The system provides a centralized platform that integrates multiple hospital functions into a single system.

2.2)User classes and characteristics

- Admin – Manages system, users, doctors and reports
- Doctor – Views patient records and updates medical data
- Receptionist – Registers patients and manages appointment
- Patients – Books appointments and views records

2.3)Assumptions

- Users have access to the internet
- The system will be hosted on a cloud platform
- Users have basic computer literacy
- The hospital operates in a private healthcare environment
- Data will be stored securely in a database

3)System features(Functional requirements)

1) User Management

FR1: The system shall allow users to register accounts.

FR2: The system shall allow users to log in using a username and password

FR3: The system shall allow users to log out securely.

2) Patient Management

FR4: The system shall allow registration of new patients.

FR5: The system shall allow updating patient details.

FR6: The system shall store patient medical history.

3) Appointment Management

FR7: The system shall allow booking appointments.

FR8: The system shall allow viewing doctor availability.

FR9: The system shall allow cancellation of appointments.

4) Doctor Management

FR10: The system shall allow the admin to add doctors.

FR11: The system shall allow the admin to remove doctors.

5) Billing System

FR12: The system shall generate invoices for patients.

FR13: The system shall process payments.

6) Inventory Management

FR14: The system shall track medical inventory.

7) Reporting

FR15: The system shall generate reports for hospital operations.

4)Non-Functional Requirements

4.1)Performance

- The system shall respond to user requests within 2 seconds.
- The system shall support at least 100 concurrent users.

4.2)Security

- The system shall encrypt user passwords.
- The system shall implement role-based access control.
- The system shall protect sensitive patient data.

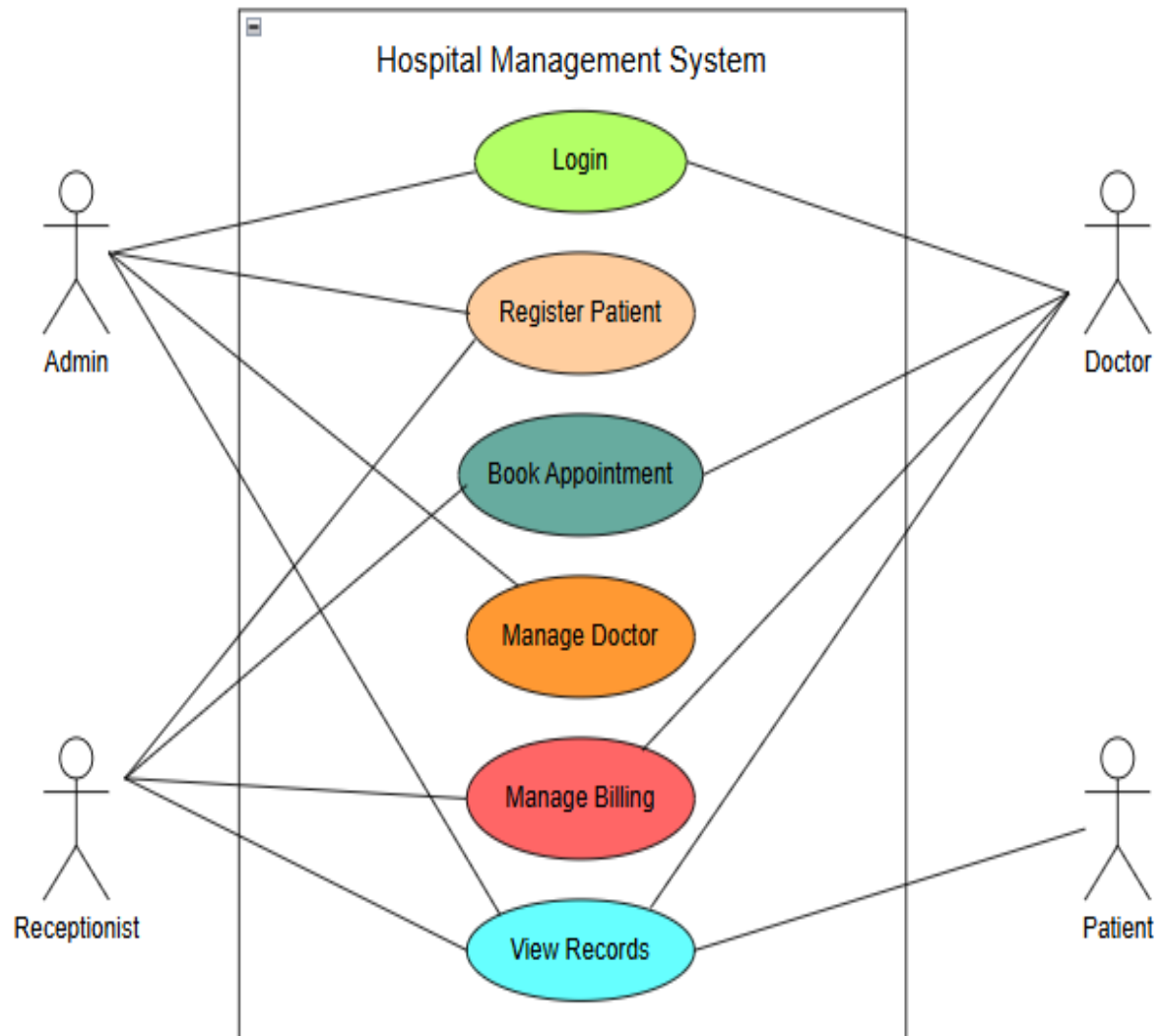
4.3)Usability

- The system shall provide a user-friendly interface.
- The system shall ensure easy navigation for all user types.

4.4)Reliability

- The system shall be available 24/7.
- The system shall ensure data consistency and prevent data loss.

5) Use Case Diagram(UML)



6)Use Case Descriptions

➤ Use Case 1: Login

- ✓ **Actor** – Admin, Doctor, Receptionist, Patient
- ✓ **Description** – User logs into the system
- ✓ **Precondition** – User must be registered
- ✓ **Postcondition** – User gains access to system
- ✓ **Main Flow** – Enter username and password → System validates

➤ Use Case 2: Book Appointment

- ✓ **Actor** – Patient / Receptionist
- ✓ **Description** – Book an appointment with a doctor
- ✓ **Precondition** – Patient must exist
- ✓ **Postcondition** – Appointment is successfully created
- ✓ **Main Flow** – Select doctor, date/time → Confirm booking

➤ Use Case 3: Manage Patient Records

- ✓ **Actor** – Doctor
- ✓ **Description** – View and update patient medical records
- ✓ **Precondition** – Patient must exist
- ✓ **Postcondition** – Patient record is updated
- ✓ **Main Flow** – Search patient → View details → Update records

7)Other Requirements

7.1)Constraints

- The system must be developed using web technologies (React, Node.js, MongoDB)
- Limited budget and development time
- Must comply with basic healthcare data handling practices

7.2)Future Enhancements

- Mobile application development
- AI-based diagnosis support
- Online payment integration
- Chatbot for patient interaction
- Multi-language support (Sinhala, Tamil, English)

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